39 · Erection, the breast, and her female words: the fine print

The words of this track, as it says them. Bold brown is the examiner.

Erection, the breast, and her female words. The fine print.

The last of the fine print: her erection steps, which she asks as put-them-in-order, the breast, from her lactation case quiz, and her two big matching tables of female words.

The hook: nitric oxide, widen, flow, fill, and drain. The breast sits on the pectoralis major. And pregnancy is forty weeks.

Her erection steps, in order. It starts with a thought or an action triggering the parasympathetic nervous system. Then nitric oxide, a vasodilator, is released. The central artery, the other arterioles, and the mesh of spongy erectile tissue widen. More blood flows through the central artery into the spongy tissue. The corpora cavernosa fill with blood and press against their outer walls, squashing the veins, so blood can't flow back out, and the penis is rigid. Last, the blood is redirected back out into the body, and the penis returns to its flaccid state. In her numbering: two, five, four, three, one.

Her male true or falses. An enlarged testicle can make it hard to urinate: false. It's an enlarged prostate that does that, because the prostate wraps around the urethra. Men have only one opening for urine and semen: true. The prostaglandins from the seminal vesicles bring on involuntary peristalsis of smooth muscle in both the male and the female tracts, helping sperm reach the fallopian tubes: true. Semen is produced by the seminal vesicles and the prostate, both. Males secrete testosterone, FSH, and LH: all three. And if testosterone doesn't work during the development of a male's brain, the brain might have a female orientation, or gender identity may be ambiguous. Both are possible.

The breast, her lactation case quiz. The breast sits directly on the pectoralis major muscle, over the intercostal muscles and the ribs, and her diagram adds the clavicle, part of the appendicular skeleton. Inside are adipose tissue, the lobules with their alveoli, which make milk under prolactin, and the lactiferous ducts, which release it under oxytocin, and open at the nipple. The areola, the dark area round the nipple, contains sebaceous glands, and darkens in pregnancy. The nipple contains erectile tissue, and hardens by smooth muscle contraction, in response to sympathetic stimuli.

Oestrogen is responsible for primary breast development, at puberty. Progesterone is responsible for further breast development, in pregnancy. Her true or false: breast development is a primary sexual characteristic. False: it's a secondary one. Another: breasts are well supported by strong ligaments and muscles. False. The breast has no muscle of its own: it hangs from its suspensory ligaments. That reason is the podcast's, but the false is her key. And her case: a woman who isn't pregnant takes high doses of oestrogen and progesterone, and then prolactin. Her mammary glands begin lactating.

Her female words, two big matching tables. Ovulation: the release of an egg from the ovary. Fertilisation: the fusion of an egg and a sperm. Contraception: a procedure or device to prevent pregnancy. Menstruation: the shedding of the endometrium each month. It's the stratum functionalis that's shed, after the spiral arteries constrict and cut off its blood. Pregnancy: the forty weeks during which a new individual develops in the mother. And abortion: the termination of a pregnancy by the removal or expulsion of an embryo or foetus.

The organs, her way. The uterus is a hollow, pear-shaped, elastic, muscular structure where a fertilised ovum can grow to a full-term baby. The ovary is a gland that makes oestrogen and oocytes. The placenta forms on the inside of the uterus, and takes oxygen and nutrients from the mother's blood to the foetus, through the umbilical cord. And one secondary sexual characteristic promoted by oestrogen, from her quiz: enlargement of the external genitalia.

The traps. The steps of an erection end with blood draining back out. An enlarged prostate, not a testicle, makes it hard to urinate. Breast development is secondary, not primary. And pregnancy is forty weeks.

Once more. Erection, the breast, and her female words.

A parasympathetic trigger, then nitric oxide is released, the central artery, arterioles and spongy tissue widen, more blood flows in, the corpora cavernosa fill and squash the veins, and last the blood drains back out: two, five, four, three, one. An enlarged prostate, not a testicle, makes it hard to urinate. Prostaglandins drive peristalsis in both tracts. Males secrete testosterone, FSH and LH. The breast sits on the pectoralis major. The alveoli make milk under prolactin, and the lactiferous ducts release it under oxytocin. The areola has sebaceous glands and darkens in pregnancy, and the nipple hardens under sympathetic stimuli.

Oestrogen gives primary breast development at puberty, and progesterone further development in pregnancy. Breast development is a secondary sexual characteristic, and the breast has no strong muscles supporting it. Ovulation releases an egg, fertilisation fuses egg and sperm, menstruation sheds the stratum functionalis, and pregnancy is forty weeks. Abortion is the termination of a pregnancy by removal or expulsion of an embryo or foetus. The uterus is hollow and pear-shaped, the ovary makes oestrogen and oocytes, and the placenta feeds the foetus through the umbilical cord. Oestrogen enlarges the external genitalia.